Comprehensive LLM Capabilities for the Project

Transformation

Converting between different formats and structures

Example: Text to JSON, markdown to structured data

01

Implementation

LLMs can transform content using structured output techniques like function calling (defining output schemas) or response formatting (XML/JSON templates)

02

Technical Detail

Libraries like LangChain's output parsers or OpenAI's function calling API enable reliable format conversion

03

Real-world Example

Notion AI converting meeting transcripts into structured action items with assignees and deadlines

PKM Relevance: Supports Collect by standardising information formats

Extraction

Pulling specific information from larger documents

Example: Extracting key facts, claims, or data points from research papers

01

Implementation

Targeted prompting with specific extraction instructions, often using "Extract all [X] from the following text"

02

Technical Detail

For complex extraction, chain-of-thought prompting significantly improves accuracy by guiding the LLM through a step-by-step reasoning process

03

Real-world Example

Elicit.ai extracting methodology details and findings from scientific papers to compare research approaches

PKM Relevance: Enhances Collect by isolating relevant information

Summarisation (Extractive and Abstractive)

Creating concise versions of longer content

Extractive: Selecting important sentences vs. **Abstractive:** Rewriting in new words

01

Implementation

For long documents, recursive summarisation breaks content into chunks, summarises each, then summarises those summaries

02

Technical Detail

Control parameters like max_tokens and temperature allow fine-tuning summary length and creativity

03

Real-world Example

Otter.ai's Meeting Notes feature that condenses hour-long meetings into key points and action items

PKM Relevance: Supports Collect and Reflect by distilling essential information

Classification & Categorisation

Automatically tagging and categorising content

Assigning documents to predefined categories

01

Implementation

Few-shot learning with examples of correctly categorised content, or zero-shot classification using predefined taxonomies

02

Technical Detail

Comparing embedding similarity scores to predefined category embeddings often outperforms prompt-only approaches

03

Real-world Example

Gmail's automatic email categorisation system that sorts messages into Primary, Social, and Promotional tabs

PKM Relevance: Enhances Collect and Surface by organising information

Semantic Search

Finding conceptually related content without exact keyword matches

Using embeddings to understand meaning, not just words

01

Implementation

Converting documents and queries into vector embeddings, then comparing their similarity in vector space

02

Technical Detail

Cosine similarity between normalised vectors is the most common comparison metric, with values closer to 1 indicating greater similarity

03

Real-world Example

GitHub Copilot's code search that finds conceptually similar code snippets even without exact keyword matches

PKM Relevance: Core to Surface by improving information findability

RAG

Responding to specific queries about your knowledge base

Drawing answers from multiple sources

01

Implementation

Retrieval-Augmented Generation (RAG) pipelines that search for relevant context, then generate answers based on retrieved information

02

Technical Detail

Hybrid retrieval combining keyword search (TF-IDF/BM25) with semantic search often outperforms either method alone

03

Real-world Example

NotebookLM which answers questions based on multiple uploaded sources

PKM Relevance: Supports Surface by providing direct answers, not just documents

Connection Discovery

Identifying relationships between concepts and documents

Building knowledge graphs automatically

01

Implementation

Entity extraction followed by relationship classification between pairs of entities

02

Technical Detail

Relationships can be stored in knowledge graphs for exploration and analysis

03

Real-world Example

Connected Papers visualisation tool showing conceptual relationships between academic publications

PKM Relevance: Essential for Connect, revealing non-obvious relationships

Text Generation & Expansion

Creating new content based on existing knowledge

Elaborating on ideas with supporting details

01

Implementation

The autoregressive nature of LLMs.

02

Technical Detail

Controlling generation characteristics through temperature and top_p parameters to balance creativity and coherence

03

Real-world Example

Chatbots such as ChatGPT. Jasper AI expanding marketing bullet points into full blog posts whilst maintaining brand voice and style

PKM Relevance: Supports Synthesise by developing ideas further

Knowledge Synthesis

Creating new knowledge from existing information

Combining multiple perspectives or sources

01

Implementation

Multi-step prompting where initial search results are processed and integrated using explicit synthesis instructions

02

Technical Detail

Map-reduce patterns first analyse individual sources (map) then combine insights (reduce) for complex synthesis tasks

03

Real-world Example

Elicit.org creating literature reviews that identify consensus views and areas of disagreement across scientific papers

PKM Relevance: Core to Synthesise mode, creating higher-level insights

Reasoning & Inference

Drawing logical conclusions from existing information

Identifying implications not explicitly stated

01

Implementation

Chain-of-thought prompting with explicit instructions to reason step-by-step

02

Technical Detail

Tree of Thoughts approaches explore multiple reasoning paths simultaneously to find the most robust conclusion

03

Real-world Example

Claude AI analysing complex logical arguments and identifying unstated assumptions or conclusions

PKM Relevance: Enhances Reflect and Synthesise with logical extensions

Text Rewriting & Adaptation

Paraphrasing and adjusting content for different purposes

Changing complexity level or tone

01

Implementation

Style transfer techniques that explicitly define target audience and communication parameters

02

Technical Detail

Explicit instructions like "Rewrite this for a [technical/non-technical] audience" with examples of desired style

03

Real-world Example

Grammarly's tone adjustment features that rewrite text to be more formal, confident, or friendly based on the communication context

PKM Relevance: Helps Surface by presenting information appropriately

Translation

Processing multilingual knowledge bases

Translating between languages

01

Implementation

Using multilingual models that can understand and generate content across dozens of languages

02

Technical Detail

Language identification followed by translation, with special handling for technical terminology and domain-specific vocabulary

03

Real-world Example

DeepL's contextual translations that preserve technical terminology and domain-specific meaning

PKM Relevance: Expands Collect and Surface across language barriers

Code Generation & Analysis

Creating code based on natural language descriptions

Explaining existing code functionality

01

Implementation

Specialised models fine-tuned on code repositories, with attention to syntax rules and programming patterns

02

Technical Detail

Abstract Syntax Tree (AST) validation ensures generated code is syntactically correct and follows language-specific rules

03

Real-world Example

Claude Code generating functional code implementations from natural language descriptions of functionality

PKM Relevance: Supports Collect and Synthesise for programming knowledge

Function Calling

Enabling LLMs to interact with external tools and APIs

Manipulating context and performing actions beyond text generation

01

Implementation

The LLM identifies user intent that requires an external action, selects the appropriate function, generates its arguments, and then processes the tool's output

02

Technical Detail

LLMs are provided with tool schemas (e.g., OpenAPI specifications) that describe available functions and their parameters, enabling structured and reliable interaction

03

Real-world Example

A customer service chatbot automatically checking order status by calling an e-commerce API based on a customer's query

PKM Relevance: Extends LLM capabilities to interact dynamically with external systems, supporting Synthesise and Reflect

Agentic Capabilities

Empowering LLMs to act autonomously, plan tasks, and execute actions to achieve defined objectives.

Moving beyond simple query-response to proactive problem-solving and task completion.

01

Implementation

Involves an iterative loop of planning, acting (using tools), observing results, and refining the plan until the objective is met.

02

Technical Detail

Leverages frameworks like CrewAI, LangChain Agents or AutoGen, which integrate LLMs with a suite of tools and a control loop for decision-making and execution.

03

Real-world Example

An LLM agent autonomously researching a topic, accessing databases, summarising findings, and generating a detailed report without continuous human input.

PKM Relevance: Automates complex, multi-step knowledge workflows, transforming information into actionable insights and supporting all PKM modes.

Multi-modal Understanding

Processing text together with images, audio, video etc.

Connecting across different content types

01

Implementation

Models that align different modalities in a shared embedding space, allowing cross-modal search and understanding

02

Technical Detail

CLIP-like architectures train on pairs of images and descriptions to learn associations between visual and textual elements

03

Real-world Example

Extracting the text content from scanned PDF documents, automatically reconstructing reading order and logical structure

PKM Relevance: Enhances Collect and Connect across media types

LLMs as Perspective Engines

Changing the POV (point-of-view)

Depending on your current goal, the available data / information / knowledge will look very different to you

01

Implementation

System prompts that establish different viewpoints or conceptual frameworks for analysis, e.g. "Analyse this product strategy from engineering, marketing, and customer perspectives"

02

Technical Detail

Set out different goals using prompts. Role-playing techniques with explicit persona definitions that include background, expertise, and typical concerns. Use different embedding strategies etc.

03

Real-world Example

A "Debate" feature that presents multiple viewpoints on controversial topics for an online newspaper

PKM Relevance: All modes.